

by_normal_form_q17 over GF(17)

June 30, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + 4X_1^4 + 2X_2^4 + 15X_0^3X_1 + 6X_0^3X_2 + 6X_0X_1^3 + 14X_1^3X_2 + 6X_0X_2^3 + 8X_1X_2^3 + 12X_0^2X_1^2 + 15X_0^2X_2^2 + 15X_1^2X_2^2 + 3X_0^2X_1X_2 + 15X_0X_1^2X_2$$

$$(1, 4, 2, 15, 6, 6, 14, 6, 8, 12, 15, 15, 3, 15, 0)$$

$$X_0^4 + 4X_1^4 + 2X_2^4 + 15X_0^3X_1 + 6X_0^3X_2 + 6X_0X_1^3 + 14X_1^3X_2 + 6X_0X_2^3 + 8X_1X_2^3$$

General information

Number of bitangents	28
Number of points	12
Fullness	is full
Number of Kovalevski points	15
Bitangent line type (a_0, a_1, a_2)	(16, 12, 0)
Number of singular points	0

All points on the curve

The quartic has 12 points:

The points on the quartic curve are:

$$\begin{array}{lll} 0 : P_{28} = (9, 0, 1) & 5 : P_{144} = (7, 7, 1) & 10 : P_{253} = (14, 13, 1) \\ 1 : P_{92} = (6, 4, 1) & 6 : P_{161} = (7, 8, 1) & 11 : P_{305} = (15, 16, 1) \\ 2 : P_{123} = (3, 6, 1) & 7 : P_{217} = (12, 11, 1) & \\ 3 : P_{135} = (15, 6, 1) & 8 : P_{227} = (5, 12, 1) & \\ 4 : P_{138} = (1, 7, 1) & 9 : P_{232} = (10, 12, 1) & \end{array}$$

The points by rank are: (28, 92, 123, 135, 138, 144, 161, 217, 227, 232, 253, 305)

The Kovalevski points are:

- 0 : $P_{127} = (7, 6, 1) = b_1 \cap b_6 \cap c_{14} \cap c_{46}$
- 1 : $P_{256} = (0, 14, 1) = a_3 \cap a_4 \cap c_{36} \cap c_{46}$
- 2 : $P_{116} = (13, 5, 1) = a_4 \cap b_5 \cap c_{45} \cap d$
- 3 : $P_{230} = (8, 12, 1) = a_2 \cap b_6 \cap c_{26} \cap d$
- 4 : $P_{204} = (16, 10, 1) = c_{12} \cap c_{14} \cap c_{25} \cap c_{45}$
- 5 : $P_{190} = (2, 10, 1) = a_4 \cap b_6 \cap c_{12} \cap c_{35}$
- 6 : $P_{188} = (0, 10, 1) = a_2 \cap a_3 \cap c_{12} \cap c_{13}$
- 7 : $P_{186} = (15, 9, 1) = c_{13} \cap c_{25} \cap c_{46} \cap d$
- 8 : $P_{77} = (8, 3, 1) = a_1 \cap a_5 \cap c_{16} \cap c_{56}$
- 9 : $P_{299} = (9, 16, 1) = b_2 \cap b_3 \cap c_{24} \cap c_{34}$
- 10 : $P_{36} = (0, 1, 1) = a_3 \cap b_5 \cap c_{14} \cap c_{26}$
- 11 : $P_{148} = (11, 7, 1) = b_1 \cap b_5 \cap c_{13} \cap c_{35}$
- 12 : $P_{277} = (4, 15, 1) = c_{25} \cap c_{26} \cap c_{35} \cap c_{36}$
- 13 : $P_{66} = (14, 2, 1) = a_6 \cap b_4 \cap c_{15} \cap c_{23}$
- 14 : $P_7 = (4, 1, 0) = a_2 \cap b_1 \cap c_{36} \cap c_{45}$

The incidence structure induced by the Kovalevski points is:

Flags=(8, 18, 21, 29, 31, 36, 40, 46, 47, 50, 68, 88, 90, 101, 104, 114, 129, 148,
152, 160, 161, 165, 168, 170, 184, 185, 186, 201, 202, 206, 210, 214, 220, 238,
248, 268, 279, 289, 292, 297, 303, 310, 312, 324, 335, 341, 342, 346, 357, 359,
362, 364, 374, 375, 376, 382, 398, 407, 408, 412)

nb_rows=28

nb_cols=15

nb_flags=60

Incma=

(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1)
(0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0)
(0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0)
(1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0)
(1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0)
(0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0)

(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0)
(0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1)
(0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
(1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0)

The Kovalevski points sorted by rank are: (7, 36, 66, 77, 116, 127, 148, 186, 188, 190, 204, 230, 256, 277, 299)

The points off the curve are:

0 : $P_0 = (1, 0, 0)$	34 : $P_{35} = (16, 0, 1)$	68 : $P_{69} = (0, 3, 1)$
1 : $P_1 = (0, 1, 0)$	35 : $P_{36} = (0, 1, 1)$	69 : $P_{70} = (1, 3, 1)$
2 : $P_2 = (0, 0, 1)$	36 : $P_{37} = (2, 1, 1)$	70 : $P_{71} = (2, 3, 1)$
3 : $P_3 = (1, 1, 1)$	37 : $P_{38} = (3, 1, 1)$	71 : $P_{72} = (3, 3, 1)$
4 : $P_4 = (1, 1, 0)$	38 : $P_{39} = (4, 1, 1)$	72 : $P_{73} = (4, 3, 1)$
5 : $P_5 = (2, 1, 0)$	39 : $P_{40} = (5, 1, 1)$	73 : $P_{74} = (5, 3, 1)$
6 : $P_6 = (3, 1, 0)$	40 : $P_{41} = (6, 1, 1)$	74 : $P_{75} = (6, 3, 1)$
7 : $P_7 = (4, 1, 0)$	41 : $P_{42} = (7, 1, 1)$	75 : $P_{76} = (7, 3, 1)$
8 : $P_8 = (5, 1, 0)$	42 : $P_{43} = (8, 1, 1)$	76 : $P_{77} = (8, 3, 1)$
9 : $P_9 = (6, 1, 0)$	43 : $P_{44} = (9, 1, 1)$	77 : $P_{78} = (9, 3, 1)$
10 : $P_{10} = (7, 1, 0)$	44 : $P_{45} = (10, 1, 1)$	78 : $P_{79} = (10, 3, 1)$
11 : $P_{11} = (8, 1, 0)$	45 : $P_{46} = (11, 1, 1)$	79 : $P_{80} = (11, 3, 1)$
12 : $P_{12} = (9, 1, 0)$	46 : $P_{47} = (12, 1, 1)$	80 : $P_{81} = (12, 3, 1)$
13 : $P_{13} = (10, 1, 0)$	47 : $P_{48} = (13, 1, 1)$	81 : $P_{82} = (13, 3, 1)$
14 : $P_{14} = (11, 1, 0)$	48 : $P_{49} = (14, 1, 1)$	82 : $P_{83} = (14, 3, 1)$
15 : $P_{15} = (12, 1, 0)$	49 : $P_{50} = (15, 1, 1)$	83 : $P_{84} = (15, 3, 1)$
16 : $P_{16} = (13, 1, 0)$	50 : $P_{51} = (16, 1, 1)$	84 : $P_{85} = (16, 3, 1)$
17 : $P_{17} = (14, 1, 0)$	51 : $P_{52} = (0, 2, 1)$	85 : $P_{86} = (0, 4, 1)$
18 : $P_{18} = (15, 1, 0)$	52 : $P_{53} = (1, 2, 1)$	86 : $P_{87} = (1, 4, 1)$
19 : $P_{19} = (16, 1, 0)$	53 : $P_{54} = (2, 2, 1)$	87 : $P_{88} = (2, 4, 1)$
20 : $P_{20} = (1, 0, 1)$	54 : $P_{55} = (3, 2, 1)$	88 : $P_{89} = (3, 4, 1)$
21 : $P_{21} = (2, 0, 1)$	55 : $P_{56} = (4, 2, 1)$	89 : $P_{90} = (4, 4, 1)$
22 : $P_{22} = (3, 0, 1)$	56 : $P_{57} = (5, 2, 1)$	90 : $P_{91} = (5, 4, 1)$
23 : $P_{23} = (4, 0, 1)$	57 : $P_{58} = (6, 2, 1)$	91 : $P_{93} = (7, 4, 1)$
24 : $P_{24} = (5, 0, 1)$	58 : $P_{59} = (7, 2, 1)$	92 : $P_{94} = (8, 4, 1)$
25 : $P_{25} = (6, 0, 1)$	59 : $P_{60} = (8, 2, 1)$	93 : $P_{95} = (9, 4, 1)$
26 : $P_{26} = (7, 0, 1)$	60 : $P_{61} = (9, 2, 1)$	94 : $P_{96} = (10, 4, 1)$
27 : $P_{27} = (8, 0, 1)$	61 : $P_{62} = (10, 2, 1)$	95 : $P_{97} = (11, 4, 1)$
28 : $P_{29} = (10, 0, 1)$	62 : $P_{63} = (11, 2, 1)$	96 : $P_{98} = (12, 4, 1)$
29 : $P_{30} = (11, 0, 1)$	63 : $P_{64} = (12, 2, 1)$	97 : $P_{99} = (13, 4, 1)$
30 : $P_{31} = (12, 0, 1)$	64 : $P_{65} = (13, 2, 1)$	98 : $P_{100} = (14, 4, 1)$
31 : $P_{32} = (13, 0, 1)$	65 : $P_{66} = (14, 2, 1)$	99 : $P_{101} = (15, 4, 1)$
32 : $P_{33} = (14, 0, 1)$	66 : $P_{67} = (15, 2, 1)$	100 : $P_{102} = (16, 4, 1)$
33 : $P_{34} = (15, 0, 1)$	67 : $P_{68} = (16, 2, 1)$	101 : $P_{103} = (0, 5, 1)$

102 : $P_{104} = (1, 5, 1)$	148 : $P_{154} = (0, 8, 1)$	194 : $P_{201} = (13, 10, 1)$
103 : $P_{105} = (2, 5, 1)$	149 : $P_{155} = (1, 8, 1)$	195 : $P_{202} = (14, 10, 1)$
104 : $P_{106} = (3, 5, 1)$	150 : $P_{156} = (2, 8, 1)$	196 : $P_{203} = (15, 10, 1)$
105 : $P_{107} = (4, 5, 1)$	151 : $P_{157} = (3, 8, 1)$	197 : $P_{204} = (16, 10, 1)$
106 : $P_{108} = (5, 5, 1)$	152 : $P_{158} = (4, 8, 1)$	198 : $P_{205} = (0, 11, 1)$
107 : $P_{109} = (6, 5, 1)$	153 : $P_{159} = (5, 8, 1)$	199 : $P_{206} = (1, 11, 1)$
108 : $P_{110} = (7, 5, 1)$	154 : $P_{160} = (6, 8, 1)$	200 : $P_{207} = (2, 11, 1)$
109 : $P_{111} = (8, 5, 1)$	155 : $P_{162} = (8, 8, 1)$	201 : $P_{208} = (3, 11, 1)$
110 : $P_{112} = (9, 5, 1)$	156 : $P_{163} = (9, 8, 1)$	202 : $P_{209} = (4, 11, 1)$
111 : $P_{113} = (10, 5, 1)$	157 : $P_{164} = (10, 8, 1)$	203 : $P_{210} = (5, 11, 1)$
112 : $P_{114} = (11, 5, 1)$	158 : $P_{165} = (11, 8, 1)$	204 : $P_{211} = (6, 11, 1)$
113 : $P_{115} = (12, 5, 1)$	159 : $P_{166} = (12, 8, 1)$	205 : $P_{212} = (7, 11, 1)$
114 : $P_{116} = (13, 5, 1)$	160 : $P_{167} = (13, 8, 1)$	206 : $P_{213} = (8, 11, 1)$
115 : $P_{117} = (14, 5, 1)$	161 : $P_{168} = (14, 8, 1)$	207 : $P_{214} = (9, 11, 1)$
116 : $P_{118} = (15, 5, 1)$	162 : $P_{169} = (15, 8, 1)$	208 : $P_{215} = (10, 11, 1)$
117 : $P_{119} = (16, 5, 1)$	163 : $P_{170} = (16, 8, 1)$	209 : $P_{216} = (11, 11, 1)$
118 : $P_{120} = (0, 6, 1)$	164 : $P_{171} = (0, 9, 1)$	210 : $P_{218} = (13, 11, 1)$
119 : $P_{121} = (1, 6, 1)$	165 : $P_{172} = (1, 9, 1)$	211 : $P_{219} = (14, 11, 1)$
120 : $P_{122} = (2, 6, 1)$	166 : $P_{173} = (2, 9, 1)$	212 : $P_{220} = (15, 11, 1)$
121 : $P_{124} = (4, 6, 1)$	167 : $P_{174} = (3, 9, 1)$	213 : $P_{221} = (16, 11, 1)$
122 : $P_{125} = (5, 6, 1)$	168 : $P_{175} = (4, 9, 1)$	214 : $P_{222} = (0, 12, 1)$
123 : $P_{126} = (6, 6, 1)$	169 : $P_{176} = (5, 9, 1)$	215 : $P_{223} = (1, 12, 1)$
124 : $P_{127} = (7, 6, 1)$	170 : $P_{177} = (6, 9, 1)$	216 : $P_{224} = (2, 12, 1)$
125 : $P_{128} = (8, 6, 1)$	171 : $P_{178} = (7, 9, 1)$	217 : $P_{225} = (3, 12, 1)$
126 : $P_{129} = (9, 6, 1)$	172 : $P_{179} = (8, 9, 1)$	218 : $P_{226} = (4, 12, 1)$
127 : $P_{130} = (10, 6, 1)$	173 : $P_{180} = (9, 9, 1)$	219 : $P_{228} = (6, 12, 1)$
128 : $P_{131} = (11, 6, 1)$	174 : $P_{181} = (10, 9, 1)$	220 : $P_{229} = (7, 12, 1)$
129 : $P_{132} = (12, 6, 1)$	175 : $P_{182} = (11, 9, 1)$	221 : $P_{230} = (8, 12, 1)$
130 : $P_{133} = (13, 6, 1)$	176 : $P_{183} = (12, 9, 1)$	222 : $P_{231} = (9, 12, 1)$
131 : $P_{134} = (14, 6, 1)$	177 : $P_{184} = (13, 9, 1)$	223 : $P_{233} = (11, 12, 1)$
132 : $P_{136} = (16, 6, 1)$	178 : $P_{185} = (14, 9, 1)$	224 : $P_{234} = (12, 12, 1)$
133 : $P_{137} = (0, 7, 1)$	179 : $P_{186} = (15, 9, 1)$	225 : $P_{235} = (13, 12, 1)$
134 : $P_{139} = (2, 7, 1)$	180 : $P_{187} = (16, 9, 1)$	226 : $P_{236} = (14, 12, 1)$
135 : $P_{140} = (3, 7, 1)$	181 : $P_{188} = (0, 10, 1)$	227 : $P_{237} = (15, 12, 1)$
136 : $P_{141} = (4, 7, 1)$	182 : $P_{189} = (1, 10, 1)$	228 : $P_{238} = (16, 12, 1)$
137 : $P_{142} = (5, 7, 1)$	183 : $P_{190} = (2, 10, 1)$	229 : $P_{239} = (0, 13, 1)$
138 : $P_{143} = (6, 7, 1)$	184 : $P_{191} = (3, 10, 1)$	230 : $P_{240} = (1, 13, 1)$
139 : $P_{145} = (8, 7, 1)$	185 : $P_{192} = (4, 10, 1)$	231 : $P_{241} = (2, 13, 1)$
140 : $P_{146} = (9, 7, 1)$	186 : $P_{193} = (5, 10, 1)$	232 : $P_{242} = (3, 13, 1)$
141 : $P_{147} = (10, 7, 1)$	187 : $P_{194} = (6, 10, 1)$	233 : $P_{243} = (4, 13, 1)$
142 : $P_{148} = (11, 7, 1)$	188 : $P_{195} = (7, 10, 1)$	234 : $P_{244} = (5, 13, 1)$
143 : $P_{149} = (12, 7, 1)$	189 : $P_{196} = (8, 10, 1)$	235 : $P_{245} = (6, 13, 1)$
144 : $P_{150} = (13, 7, 1)$	190 : $P_{197} = (9, 10, 1)$	236 : $P_{246} = (7, 13, 1)$
145 : $P_{151} = (14, 7, 1)$	191 : $P_{198} = (10, 10, 1)$	237 : $P_{247} = (8, 13, 1)$
146 : $P_{152} = (15, 7, 1)$	192 : $P_{199} = (11, 10, 1)$	238 : $P_{248} = (9, 13, 1)$
147 : $P_{153} = (16, 7, 1)$	193 : $P_{200} = (12, 10, 1)$	239 : $P_{249} = (10, 13, 1)$

240 : $P_{250} = (11, 13, 1)$	259 : $P_{270} = (14, 14, 1)$	278 : $P_{289} = (16, 15, 1)$
241 : $P_{251} = (12, 13, 1)$	260 : $P_{271} = (15, 14, 1)$	279 : $P_{290} = (0, 16, 1)$
242 : $P_{252} = (13, 13, 1)$	261 : $P_{272} = (16, 14, 1)$	280 : $P_{291} = (1, 16, 1)$
243 : $P_{254} = (15, 13, 1)$	262 : $P_{273} = (0, 15, 1)$	281 : $P_{292} = (2, 16, 1)$
244 : $P_{255} = (16, 13, 1)$	263 : $P_{274} = (1, 15, 1)$	282 : $P_{293} = (3, 16, 1)$
245 : $P_{256} = (0, 14, 1)$	264 : $P_{275} = (2, 15, 1)$	283 : $P_{294} = (4, 16, 1)$
246 : $P_{257} = (1, 14, 1)$	265 : $P_{276} = (3, 15, 1)$	284 : $P_{295} = (5, 16, 1)$
247 : $P_{258} = (2, 14, 1)$	266 : $P_{277} = (4, 15, 1)$	285 : $P_{296} = (6, 16, 1)$
248 : $P_{259} = (3, 14, 1)$	267 : $P_{278} = (5, 15, 1)$	286 : $P_{297} = (7, 16, 1)$
249 : $P_{260} = (4, 14, 1)$	268 : $P_{279} = (6, 15, 1)$	287 : $P_{298} = (8, 16, 1)$
250 : $P_{261} = (5, 14, 1)$	269 : $P_{280} = (7, 15, 1)$	288 : $P_{299} = (9, 16, 1)$
251 : $P_{262} = (6, 14, 1)$	270 : $P_{281} = (8, 15, 1)$	289 : $P_{300} = (10, 16, 1)$
252 : $P_{263} = (7, 14, 1)$	271 : $P_{282} = (9, 15, 1)$	290 : $P_{301} = (11, 16, 1)$
253 : $P_{264} = (8, 14, 1)$	272 : $P_{283} = (10, 15, 1)$	291 : $P_{302} = (12, 16, 1)$
254 : $P_{265} = (9, 14, 1)$	273 : $P_{284} = (11, 15, 1)$	292 : $P_{303} = (13, 16, 1)$
255 : $P_{266} = (10, 14, 1)$	274 : $P_{285} = (12, 15, 1)$	293 : $P_{304} = (14, 16, 1)$
256 : $P_{267} = (11, 14, 1)$	275 : $P_{286} = (13, 15, 1)$	294 : $P_{306} = (16, 16, 1)$
257 : $P_{268} = (12, 14, 1)$	276 : $P_{287} = (14, 15, 1)$	
258 : $P_{269} = (13, 14, 1)$	277 : $P_{288} = (15, 15, 1)$	

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 136, 137, 139, 140, 141, 142, 143, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 218, 219, 220, 221, 222, 223, 224, 225, 226, 228, 229, 230, 231, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 306)

Duals of Bitangents

dual of bitangents:

- 0 : $P_{95} = (9, 4, 1)$
- 1 : $P_{106} = (3, 5, 1)$
- 2 : $P_0 = (1, 0, 0)$
- 3 : $P_{132} = (12, 6, 1)$

4 : $P_{170} = (16, 8, 1)$
 5 : $P_{25} = (6, 0, 1)$
 6 : $P_7 = (4, 1, 0)$
 7 : $P_{36} = (0, 1, 1)$
 8 : $P_{54} = (2, 2, 1)$
 9 : $P_{207} = (2, 11, 1)$
 10 : $P_{306} = (16, 16, 1)$
 11 : $P_{185} = (14, 9, 1)$
 12 : $P_{103} = (0, 5, 1)$
 13 : $P_{109} = (6, 5, 1)$
 14 : $P_{298} = (8, 16, 1)$
 15 : $P_{142} = (5, 7, 1)$
 16 : $P_{228} = (6, 12, 1)$
 17 : $P_{89} = (3, 4, 1)$
 18 : $P_{227} = (5, 12, 1)$
 19 : $P_{83} = (14, 3, 1)$
 20 : $P_{302} = (12, 16, 1)$
 21 : $P_{34} = (15, 0, 1)$
 22 : $P_{197} = (9, 10, 1)$
 23 : $P_{127} = (7, 6, 1)$
 24 : $P_{261} = (5, 14, 1)$
 25 : $P_{122} = (2, 6, 1)$
 26 : $P_{280} = (7, 15, 1)$
 27 : $P_{247} = (8, 13, 1)$
 Dimension of the vanishing ideal is 0

The lines and their points of contact are:

$$\begin{aligned}
 a_1 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 13 \end{bmatrix}_{157} P_{217} = \mathbf{P}(12, 11, 1) \ 4\times \\
 a_2 &= \begin{bmatrix} 1 & 0 & 14 \\ 0 & 1 & 12 \end{bmatrix}_{264} \\
 a_3 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{306} \\
 a_4 &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 11 \end{bmatrix}_{101} \\
 a_5 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 9 \end{bmatrix}_{27} P_{135} = \mathbf{P}(15, 6, 1) \ 4\times \\
 a_6 &= \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 0 \end{bmatrix}_{198} P_{253} = \mathbf{P}(14, 13, 1) \ 4\times \\
 b_1 &= \begin{bmatrix} 1 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{251} \\
 b_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 16 \end{bmatrix}_{16} P_{305} = \mathbf{P}(15, 16, 1) \ 4\times
 \end{aligned}$$

$$\begin{aligned}
b_3 &= \begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & 15 \end{bmatrix}_{285} & P_{138} &= \mathbf{P}(1, 7, 1) \ 4\times \\
b_4 &= \begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & 6 \end{bmatrix}_{276} & P_{232} &= \mathbf{P}(10, 12, 1) \ 4\times \\
b_5 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{19} \\
b_6 &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 8 \end{bmatrix}_{62} \\
c_{12} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 12 \end{bmatrix}_{12} \\
c_{13} &= \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 12 \end{bmatrix}_{210} \\
c_{14} &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 1 \end{bmatrix}_{163} \\
c_{15} &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 10 \end{bmatrix}_{226} & P_{144} &= \mathbf{P}(7, 7, 1) \ 4\times \\
c_{16} &= \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 5 \end{bmatrix}_{203} & P_{92} &= \mathbf{P}(6, 4, 1) \ 4\times \\
c_{23} &= \begin{bmatrix} 1 & 0 & 14 \\ 0 & 1 & 13 \end{bmatrix}_{265} & P_{123} &= \mathbf{P}(3, 6, 1) \ 4\times \\
c_{24} &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 5 \end{bmatrix}_{221} & P_{227} &= \mathbf{P}(5, 12, 1) \ 4\times \\
c_{25} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 14 \end{bmatrix}_{68} \\
c_{26} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}_{91} \\
c_{34} &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}_{36} & P_{28} &= \mathbf{P}(9, 0, 1) \ 4\times \\
c_{35} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}_{151} \\
c_{36} &= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 11 \end{bmatrix}_{191} \\
c_{45} &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 3 \end{bmatrix}_{219} \\
c_{46} &= \begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & 11 \end{bmatrix}_{281} \\
c_{56} &= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 2 \end{bmatrix}_{182} & P_{161} &= \mathbf{P}(7, 8, 1) \ 4\times \\
d &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 4 \end{bmatrix}_{166}
\end{aligned}$$

Rank of lines: (157, 264, 306, 101, 27, 198, 251, 16, 285, 276, 19, 62, 12, 210, 163, 226, 203, 265, 221, 68, 91, 36, 151, 191, 219, 281, 182, 166)
Line type: $1^{12}, 0^{16}$

12	1	26, 5, 21, 9, 4, 18, 17, 8, 16, 15, 7, 0
16	0	19, 12, 23, 20, 25, 6, 27, 24, 22, 10, 11, 2, 13, 14, 3, 1

point types: 1^{12}

12	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 0
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point types for points off the curve: 4^{15} , 3^{48} , 2^{144} , 0^{88}

15	4	124, 245, 114, 221, 197, 183, 181, 179, 76, 288, 35, 142, 266, 65, 7
48	3	126, 242, 120, 235, 234, 28, 27, 225, 224, 13, 110, 211, 104, 207, 202, 98, 194, 190, 46, 184, 45, 180, 175, 43, 86, 85, 41, 162, 161, 156, 293, 145, 289, 287, 285, 284, 281, 280, 276, 271, 270, 133, 32, 264, 261, 256, 1, 294
144	2	141, 248, 30, 29, 241, 239, 119, 240, 231, 115, 229, 282, 59, 238, 116, 12, 108, 215, 107, 53, 219, 26, 111, 6, 213, 230, 232, 57, 105, 118, 52, 25, 121, 60, 208, 61, 252, 253, 51, 2, 122, 205, 101, 50, 201, 49, 200, 48, 97, 5, 193, 192, 191, 95, 11, 249, 189, 93, 254, 186, 185, 21, 89, 88, 176, 4, 143, 173, 84, 83, 20, 260, 166, 165, 82, 164, 81, 39, 80, 19, 169, 160, 157, 77, 155, 38, 292, 154, 177, 178, 151, 75, 90, 182, 18, 150, 149, 74, 187, 188, 148, 94, 73, 291, 144, 71, 195, 196, 17, 198, 132, 283, 140, 100, 203, 279, 138, 68, 0, 136, 272, 67, 33, 212, 268, 214, 66, 265, 131, 218, 263, 220, 54, 222, 262, 259, 112, 226, 129, 290, 63, 286, 137, 250
88	0	278, 258, 172, 72, 128, 91, 204, 64, 158, 147, 228, 31, 275, 8, 217, 102, 255, 174, 40, 159, 153, 36, 274, 55, 117, 237, 134, 24, 233, 62, 56, 109, 209, 206, 199, 70, 22, 87, 171, 168, 146, 79, 44, 37, 152, 96, 267, 135, 106, 69, 34, 236, 113, 58, 14, 277, 216, 123, 273, 139, 47, 3, 269, 243, 227, 223, 246, 210, 247, 103, 16, 99, 23, 244, 127, 92, 10, 257, 9, 251, 170, 167, 163, 15, 42, 130, 125, 78

Lines on points off the curve:

Off point $0 = P_0 = (1, 0, 0)$ lies on 2 bisecants : $\{ 7, 12 \}$

Off point 1 = $P_1 = (0, 1, 0)$ lies on 3 bisecants : $\{ 2, 5, 21 \}$
 Off point 2 = $P_2 = (0, 0, 1)$ lies on 2 bisecants : $\{ 2, 6 \}$
 Off point 3 = $P_3 = (1, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 4 = $P_4 = (1, 1, 0)$ lies on 2 bisecants : $\{ 18, 19 \}$
 Off point 5 = $P_5 = (2, 1, 0)$ lies on 2 bisecants : $\{ 13, 15 \}$
 Off point 6 = $P_6 = (3, 1, 0)$ lies on 2 bisecants : $\{ 9, 11 \}$
 Off point 7 = $P_7 = (4, 1, 0)$ lies on 4 bisecants : $\{ 1, 6, 23, 24 \}$
 Off point 8 = $P_8 = (5, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 9 = $P_9 = (6, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 10 = $P_{10} = (7, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 11 = $P_{11} = (8, 1, 0)$ lies on 2 bisecants : $\{ 3, 4 \}$
 Off point 12 = $P_{12} = (9, 1, 0)$ lies on 2 bisecants : $\{ 0, 27 \}$
 Off point 13 = $P_{13} = (10, 1, 0)$ lies on 3 bisecants : $\{ 17, 20, 26 \}$
 Off point 14 = $P_{14} = (11, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 15 = $P_{15} = (12, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 16 = $P_{16} = (13, 1, 0)$ lies on 0 bisecants : $\{ \}$
 Off point 17 = $P_{17} = (14, 1, 0)$ lies on 2 bisecants : $\{ 22, 25 \}$
 Off point 18 = $P_{18} = (15, 1, 0)$ lies on 2 bisecants : $\{ 14, 16 \}$
 Off point 19 = $P_{19} = (16, 1, 0)$ lies on 2 bisecants : $\{ 8, 10 \}$
 Off point 20 = $P_{20} = (1, 0, 1)$ lies on 2 bisecants : $\{ 4, 10 \}$
 Off point 21 = $P_{21} = (2, 0, 1)$ lies on 2 bisecants : $\{ 14, 27 \}$
 Off point 22 = $P_{22} = (3, 0, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 23 = $P_{23} = (4, 0, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 24 = $P_{24} = (5, 0, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 25 = $P_{25} = (6, 0, 1)$ lies on 2 bisecants : $\{ 11, 19 \}$
 Off point 26 = $P_{26} = (7, 0, 1)$ lies on 2 bisecants : $\{ 3, 20 \}$
 Off point 27 = $P_{27} = (8, 0, 1)$ lies on 3 bisecants : $\{ 8, 9, 25 \}$
 Off point 28 = $P_{28} = (10, 0, 1)$ lies on 3 bisecants : $\{ 15, 18, 24 \}$
 Off point 29 = $P_{29} = (11, 0, 1)$ lies on 2 bisecants : $\{ 1, 17 \}$
 Off point 30 = $P_{30} = (12, 0, 1)$ lies on 2 bisecants : $\{ 23, 26 \}$
 Off point 31 = $P_{31} = (13, 0, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 32 = $P_{32} = (14, 0, 1)$ lies on 3 bisecants : $\{ 5, 13, 16 \}$
 Off point 33 = $P_{33} = (15, 0, 1)$ lies on 2 bisecants : $\{ 0, 22 \}$
 Off point 34 = $P_{34} = (16, 0, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 35 = $P_{35} = (0, 1, 1)$ lies on 4 bisecants : $\{ 2, 10, 14, 20 \}$
 Off point 36 = $P_{36} = (2, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 37 = $P_{37} = (3, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 38 = $P_{38} = (4, 1, 1)$ lies on 2 bisecants : $\{ 6, 17 \}$
 Off point 39 = $P_{39} = (5, 1, 1)$ lies on 2 bisecants : $\{ 25, 26 \}$
 Off point 40 = $P_{40} = (6, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 41 = $P_{41} = (7, 1, 1)$ lies on 3 bisecants : $\{ 0, 8, 19 \}$
 Off point 42 = $P_{42} = (8, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 43 = $P_{43} = (9, 1, 1)$ lies on 3 bisecants : $\{ 4, 11, 21 \}$
 Off point 44 = $P_{44} = (10, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 45 = $P_{45} = (11, 1, 1)$ lies on 3 bisecants : $\{ 9, 18, 27 \}$
 Off point 46 = $P_{46} = (12, 1, 1)$ lies on 3 bisecants : $\{ 15, 16, 22 \}$

Off point 47 = $P_{48} = (13, 1, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 48 = $P_{49} = (14, 1, 1)$ lies on 2 bisecants : $\{ 5, 24 \}$
 Off point 49 = $P_{50} = (15, 1, 1)$ lies on 2 bisecants : $\{ 1, 3 \}$
 Off point 50 = $P_{51} = (16, 1, 1)$ lies on 2 bisecants : $\{ 13, 23 \}$
 Off point 51 = $P_{52} = (0, 2, 1)$ lies on 2 bisecants : $\{ 2, 4 \}$
 Off point 52 = $P_{53} = (1, 2, 1)$ lies on 2 bisecants : $\{ 13, 24 \}$
 Off point 53 = $P_{54} = (2, 2, 1)$ lies on 2 bisecants : $\{ 1, 25 \}$
 Off point 54 = $P_{55} = (3, 2, 1)$ lies on 2 bisecants : $\{ 23, 27 \}$
 Off point 55 = $P_{56} = (4, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 56 = $P_{57} = (5, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 57 = $P_{58} = (6, 2, 1)$ lies on 2 bisecants : $\{ 3, 8 \}$
 Off point 58 = $P_{59} = (7, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 59 = $P_{60} = (8, 2, 1)$ lies on 2 bisecants : $\{ 6, 19 \}$
 Off point 60 = $P_{61} = (9, 2, 1)$ lies on 2 bisecants : $\{ 21, 22 \}$
 Off point 61 = $P_{62} = (10, 2, 1)$ lies on 2 bisecants : $\{ 16, 20 \}$
 Off point 62 = $P_{63} = (11, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 63 = $P_{64} = (12, 2, 1)$ lies on 2 bisecants : $\{ 11, 18 \}$
 Off point 64 = $P_{65} = (13, 2, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 65 = $P_{66} = (14, 2, 1)$ lies on 4 bisecants : $\{ 5, 9, 15, 17 \}$
 Off point 66 = $P_{67} = (15, 2, 1)$ lies on 2 bisecants : $\{ 14, 26 \}$
 Off point 67 = $P_{68} = (16, 2, 1)$ lies on 2 bisecants : $\{ 0, 10 \}$
 Off point 68 = $P_{69} = (0, 3, 1)$ lies on 2 bisecants : $\{ 2, 9 \}$
 Off point 69 = $P_{70} = (1, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 70 = $P_{71} = (2, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 71 = $P_{72} = (3, 3, 1)$ lies on 2 bisecants : $\{ 13, 20 \}$
 Off point 72 = $P_{73} = (4, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 73 = $P_{74} = (5, 3, 1)$ lies on 2 bisecants : $\{ 8, 24 \}$
 Off point 74 = $P_{75} = (6, 3, 1)$ lies on 2 bisecants : $\{ 1, 22 \}$
 Off point 75 = $P_{76} = (7, 3, 1)$ lies on 2 bisecants : $\{ 17, 23 \}$
 Off point 76 = $P_{77} = (8, 3, 1)$ lies on 4 bisecants : $\{ 0, 4, 16, 26 \}$
 Off point 77 = $P_{78} = (9, 3, 1)$ lies on 2 bisecants : $\{ 19, 21 \}$
 Off point 78 = $P_{79} = (10, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 79 = $P_{80} = (11, 3, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 80 = $P_{81} = (12, 3, 1)$ lies on 2 bisecants : $\{ 6, 27 \}$
 Off point 81 = $P_{82} = (13, 3, 1)$ lies on 2 bisecants : $\{ 14, 18 \}$
 Off point 82 = $P_{83} = (14, 3, 1)$ lies on 2 bisecants : $\{ 3, 5 \}$
 Off point 83 = $P_{84} = (15, 3, 1)$ lies on 2 bisecants : $\{ 10, 11 \}$
 Off point 84 = $P_{85} = (16, 3, 1)$ lies on 2 bisecants : $\{ 15, 25 \}$
 Off point 85 = $P_{86} = (0, 4, 1)$ lies on 3 bisecants : $\{ 0, 2, 17 \}$
 Off point 86 = $P_{87} = (1, 4, 1)$ lies on 3 bisecants : $\{ 11, 15, 26 \}$
 Off point 87 = $P_{88} = (2, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 88 = $P_{89} = (3, 4, 1)$ lies on 2 bisecants : $\{ 9, 22 \}$
 Off point 89 = $P_{90} = (4, 4, 1)$ lies on 2 bisecants : $\{ 8, 27 \}$
 Off point 90 = $P_{91} = (5, 4, 1)$ lies on 2 bisecants : $\{ 3, 13 \}$
 Off point 91 = $P_{93} = (7, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 92 = $P_{94} = (8, 4, 1)$ lies on 0 bisecants : $\{ \}$

Off point 93 = $P_{95} = (9, 4, 1)$ lies on 2 bisecants : { 21, 24 }
 Off point 94 = $P_{96} = (10, 4, 1)$ lies on 2 bisecants : { 1, 19 }
 Off point 95 = $P_{97} = (11, 4, 1)$ lies on 2 bisecants : { 14, 23 }
 Off point 96 = $P_{98} = (12, 4, 1)$ lies on 0 bisecants : { }
 Off point 97 = $P_{99} = (13, 4, 1)$ lies on 2 bisecants : { 20, 25 }
 Off point 98 = $P_{100} = (14, 4, 1)$ lies on 3 bisecants : { 5, 10, 18 }
 Off point 99 = $P_{101} = (15, 4, 1)$ lies on 0 bisecants : { }
 Off point 100 = $P_{102} = (16, 4, 1)$ lies on 2 bisecants : { 4, 6 }
 Off point 101 = $P_{103} = (0, 5, 1)$ lies on 2 bisecants : { 2, 22 }
 Off point 102 = $P_{104} = (1, 5, 1)$ lies on 0 bisecants : { }
 Off point 103 = $P_{105} = (2, 5, 1)$ lies on 0 bisecants : { }
 Off point 104 = $P_{106} = (3, 5, 1)$ lies on 3 bisecants : { 6, 8, 15 }
 Off point 105 = $P_{107} = (4, 5, 1)$ lies on 2 bisecants : { 11, 16 }
 Off point 106 = $P_{108} = (5, 5, 1)$ lies on 0 bisecants : { }
 Off point 107 = $P_{109} = (6, 5, 1)$ lies on 2 bisecants : { 9, 20 }
 Off point 108 = $P_{110} = (7, 5, 1)$ lies on 2 bisecants : { 4, 13 }
 Off point 109 = $P_{111} = (8, 5, 1)$ lies on 0 bisecants : { }
 Off point 110 = $P_{112} = (9, 5, 1)$ lies on 3 bisecants : { 0, 14, 21 }
 Off point 111 = $P_{113} = (10, 5, 1)$ lies on 2 bisecants : { 17, 25 }
 Off point 112 = $P_{114} = (11, 5, 1)$ lies on 2 bisecants : { 19, 26 }
 Off point 113 = $P_{115} = (12, 5, 1)$ lies on 0 bisecants : { }
 Off point 114 = $P_{116} = (13, 5, 1)$ lies on 4 bisecants : { 3, 10, 24, 27 }
 Off point 115 = $P_{117} = (14, 5, 1)$ lies on 2 bisecants : { 1, 5 }
 Off point 116 = $P_{118} = (15, 5, 1)$ lies on 2 bisecants : { 18, 23 }
 Off point 117 = $P_{119} = (16, 5, 1)$ lies on 0 bisecants : { }
 Off point 118 = $P_{120} = (0, 6, 1)$ lies on 2 bisecants : { 2, 24 }
 Off point 119 = $P_{121} = (1, 6, 1)$ lies on 2 bisecants : { 0, 1 }
 Off point 120 = $P_{122} = (2, 6, 1)$ lies on 3 bisecants : { 8, 16, 23 }
 Off point 121 = $P_{124} = (4, 6, 1)$ lies on 2 bisecants : { 3, 26 }
 Off point 122 = $P_{125} = (5, 6, 1)$ lies on 2 bisecants : { 15, 27 }
 Off point 123 = $P_{126} = (6, 6, 1)$ lies on 0 bisecants : { }
 Off point 124 = $P_{127} = (7, 6, 1)$ lies on 4 bisecants : { 6, 11, 14, 25 }
 Off point 125 = $P_{128} = (8, 6, 1)$ lies on 0 bisecants : { }
 Off point 126 = $P_{129} = (9, 6, 1)$ lies on 3 bisecants : { 9, 13, 21 }
 Off point 127 = $P_{130} = (10, 6, 1)$ lies on 0 bisecants : { }
 Off point 128 = $P_{131} = (11, 6, 1)$ lies on 0 bisecants : { }
 Off point 129 = $P_{132} = (12, 6, 1)$ lies on 2 bisecants : { 10, 19 }
 Off point 130 = $P_{133} = (13, 6, 1)$ lies on 0 bisecants : { }
 Off point 131 = $P_{134} = (14, 6, 1)$ lies on 2 bisecants : { 5, 22 }
 Off point 132 = $P_{136} = (16, 6, 1)$ lies on 2 bisecants : { 18, 20 }
 Off point 133 = $P_{137} = (0, 7, 1)$ lies on 3 bisecants : { 2, 16, 18 }
 Off point 134 = $P_{139} = (2, 7, 1)$ lies on 0 bisecants : { }
 Off point 135 = $P_{140} = (3, 7, 1)$ lies on 0 bisecants : { }
 Off point 136 = $P_{141} = (4, 7, 1)$ lies on 2 bisecants : { 24, 25 }
 Off point 137 = $P_{142} = (5, 7, 1)$ lies on 2 bisecants : { 1, 14 }
 Off point 138 = $P_{143} = (6, 7, 1)$ lies on 2 bisecants : { 4, 23 }

Off point 139 = $P_{145} = (8, 7, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 140 = $P_{146} = (9, 7, 1)$ lies on 2 bisecants : $\{ 20, 21 \}$
 Off point 141 = $P_{147} = (10, 7, 1)$ lies on 2 bisecants : $\{ 0, 11 \}$
 Off point 142 = $P_{148} = (11, 7, 1)$ lies on 4 bisecants : $\{ 6, 10, 13, 22 \}$
 Off point 143 = $P_{149} = (12, 7, 1)$ lies on 2 bisecants : $\{ 3, 9 \}$
 Off point 144 = $P_{150} = (13, 7, 1)$ lies on 2 bisecants : $\{ 17, 19 \}$
 Off point 145 = $P_{151} = (14, 7, 1)$ lies on 3 bisecants : $\{ 5, 26, 27 \}$
 Off point 146 = $P_{152} = (15, 7, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 147 = $P_{153} = (16, 7, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 148 = $P_{154} = (0, 8, 1)$ lies on 2 bisecants : $\{ 2, 8 \}$
 Off point 149 = $P_{155} = (1, 8, 1)$ lies on 2 bisecants : $\{ 18, 25 \}$
 Off point 150 = $P_{156} = (2, 8, 1)$ lies on 2 bisecants : $\{ 0, 20 \}$
 Off point 151 = $P_{157} = (3, 8, 1)$ lies on 2 bisecants : $\{ 3, 14 \}$
 Off point 152 = $P_{158} = (4, 8, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 153 = $P_{159} = (5, 8, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 154 = $P_{160} = (6, 8, 1)$ lies on 2 bisecants : $\{ 17, 27 \}$
 Off point 155 = $P_{162} = (8, 8, 1)$ lies on 2 bisecants : $\{ 22, 24 \}$
 Off point 156 = $P_{163} = (9, 8, 1)$ lies on 3 bisecants : $\{ 1, 15, 21 \}$
 Off point 157 = $P_{164} = (10, 8, 1)$ lies on 2 bisecants : $\{ 10, 23 \}$
 Off point 158 = $P_{165} = (11, 8, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 159 = $P_{166} = (12, 8, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 160 = $P_{167} = (13, 8, 1)$ lies on 2 bisecants : $\{ 11, 13 \}$
 Off point 161 = $P_{168} = (14, 8, 1)$ lies on 3 bisecants : $\{ 4, 5, 19 \}$
 Off point 162 = $P_{169} = (15, 8, 1)$ lies on 3 bisecants : $\{ 6, 9, 16 \}$
 Off point 163 = $P_{170} = (16, 8, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 164 = $P_{171} = (0, 9, 1)$ lies on 2 bisecants : $\{ 2, 26 \}$
 Off point 165 = $P_{172} = (1, 9, 1)$ lies on 2 bisecants : $\{ 9, 14 \}$
 Off point 166 = $P_{173} = (2, 9, 1)$ lies on 2 bisecants : $\{ 6, 18 \}$
 Off point 167 = $P_{174} = (3, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 168 = $P_{175} = (4, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 169 = $P_{176} = (5, 9, 1)$ lies on 2 bisecants : $\{ 4, 22 \}$
 Off point 170 = $P_{177} = (6, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 171 = $P_{178} = (7, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 172 = $P_{179} = (8, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 173 = $P_{180} = (9, 9, 1)$ lies on 2 bisecants : $\{ 10, 21 \}$
 Off point 174 = $P_{181} = (10, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 175 = $P_{182} = (11, 9, 1)$ lies on 3 bisecants : $\{ 0, 3, 15 \}$
 Off point 176 = $P_{183} = (12, 9, 1)$ lies on 2 bisecants : $\{ 20, 24 \}$
 Off point 177 = $P_{184} = (13, 9, 1)$ lies on 2 bisecants : $\{ 1, 16 \}$
 Off point 178 = $P_{185} = (14, 9, 1)$ lies on 2 bisecants : $\{ 5, 23 \}$
 Off point 179 = $P_{186} = (15, 9, 1)$ lies on 4 bisecants : $\{ 13, 19, 25, 27 \}$
 Off point 180 = $P_{187} = (16, 9, 1)$ lies on 3 bisecants : $\{ 8, 11, 17 \}$
 Off point 181 = $P_{188} = (0, 10, 1)$ lies on 4 bisecants : $\{ 1, 2, 12, 13 \}$
 Off point 182 = $P_{189} = (1, 10, 1)$ lies on 2 bisecants : $\{ 12, 23 \}$
 Off point 183 = $P_{190} = (2, 10, 1)$ lies on 4 bisecants : $\{ 3, 11, 12, 22 \}$
 Off point 184 = $P_{191} = (3, 10, 1)$ lies on 3 bisecants : $\{ 0, 12, 18 \}$

Off point 185 = P_{192} = (4, 10, 1) lies on 2 bisecants : { 9, 12 }
 Off point 186 = P_{193} = (5, 10, 1) lies on 2 bisecants : { 12, 20 }
 Off point 187 = P_{194} = (6, 10, 1) lies on 2 bisecants : { 6, 12 }
 Off point 188 = P_{195} = (7, 10, 1) lies on 2 bisecants : { 12, 27 }
 Off point 189 = P_{196} = (8, 10, 1) lies on 2 bisecants : { 10, 12 }
 Off point 190 = P_{197} = (9, 10, 1) lies on 3 bisecants : { 12, 17, 21 }
 Off point 191 = P_{198} = (10, 10, 1) lies on 2 bisecants : { 12, 26 }
 Off point 192 = P_{199} = (11, 10, 1) lies on 2 bisecants : { 12, 16 }
 Off point 193 = P_{200} = (12, 10, 1) lies on 2 bisecants : { 12, 25 }
 Off point 194 = P_{201} = (13, 10, 1) lies on 3 bisecants : { 4, 12, 15 }
 Off point 195 = P_{202} = (14, 10, 1) lies on 2 bisecants : { 5, 12 }
 Off point 196 = P_{203} = (15, 10, 1) lies on 2 bisecants : { 8, 12 }
 Off point 197 = P_{204} = (16, 10, 1) lies on 4 bisecants : { 12, 14, 19, 24 }
 Off point 198 = P_{205} = (0, 11, 1) lies on 2 bisecants : { 2, 19 }
 Off point 199 = P_{206} = (1, 11, 1) lies on 0 bisecants : { }
 Off point 200 = P_{207} = (2, 11, 1) lies on 2 bisecants : { 13, 17 }
 Off point 201 = P_{208} = (3, 11, 1) lies on 2 bisecants : { 24, 26 }
 Off point 202 = P_{209} = (4, 11, 1) lies on 3 bisecants : { 1, 4, 18 }
 Off point 203 = P_{210} = (5, 11, 1) lies on 2 bisecants : { 11, 23 }
 Off point 204 = P_{211} = (6, 11, 1) lies on 0 bisecants : { }
 Off point 205 = P_{212} = (7, 11, 1) lies on 2 bisecants : { 9, 10 }
 Off point 206 = P_{213} = (8, 11, 1) lies on 0 bisecants : { }
 Off point 207 = P_{214} = (9, 11, 1) lies on 3 bisecants : { 16, 21, 25 }
 Off point 208 = P_{215} = (10, 11, 1) lies on 2 bisecants : { 3, 6 }
 Off point 209 = P_{216} = (11, 11, 1) lies on 0 bisecants : { }
 Off point 210 = P_{218} = (13, 11, 1) lies on 0 bisecants : { }
 Off point 211 = P_{219} = (14, 11, 1) lies on 3 bisecants : { 5, 8, 14 }
 Off point 212 = P_{220} = (15, 11, 1) lies on 2 bisecants : { 15, 20 }
 Off point 213 = P_{221} = (16, 11, 1) lies on 2 bisecants : { 22, 27 }
 Off point 214 = P_{222} = (0, 12, 1) lies on 2 bisecants : { 2, 15 }
 Off point 215 = P_{223} = (1, 12, 1) lies on 2 bisecants : { 3, 19 }
 Off point 216 = P_{224} = (2, 12, 1) lies on 0 bisecants : { }
 Off point 217 = P_{225} = (3, 12, 1) lies on 0 bisecants : { }
 Off point 218 = P_{226} = (4, 12, 1) lies on 2 bisecants : { 0, 13 }
 Off point 219 = P_{228} = (6, 12, 1) lies on 2 bisecants : { 10, 25 }
 Off point 220 = P_{229} = (7, 12, 1) lies on 2 bisecants : { 16, 24 }
 Off point 221 = P_{230} = (8, 12, 1) lies on 4 bisecants : { 1, 11, 20, 27 }
 Off point 222 = P_{231} = (9, 12, 1) lies on 2 bisecants : { 21, 23 }
 Off point 223 = P_{233} = (11, 12, 1) lies on 0 bisecants : { }
 Off point 224 = P_{234} = (12, 12, 1) lies on 3 bisecants : { 4, 14, 17 }
 Off point 225 = P_{235} = (13, 12, 1) lies on 3 bisecants : { 8, 22, 26 }
 Off point 226 = P_{236} = (14, 12, 1) lies on 2 bisecants : { 5, 6 }
 Off point 227 = P_{237} = (15, 12, 1) lies on 0 bisecants : { }
 Off point 228 = P_{238} = (16, 12, 1) lies on 0 bisecants : { }
 Off point 229 = P_{239} = (0, 13, 1) lies on 2 bisecants : { 2, 27 }
 Off point 230 = P_{240} = (1, 13, 1) lies on 2 bisecants : { 6, 20 }

Off point 231 = P_{241} = (2, 13, 1) lies on 2 bisecants : { 15, 19 }
 Off point 232 = P_{242} = (3, 13, 1) lies on 2 bisecants : { 4, 25 }
 Off point 233 = P_{243} = (4, 13, 1) lies on 0 bisecants : { }
 Off point 234 = P_{244} = (5, 13, 1) lies on 3 bisecants : { 10, 16, 17 }
 Off point 235 = P_{245} = (6, 13, 1) lies on 3 bisecants : { 13, 18, 26 }
 Off point 236 = P_{246} = (7, 13, 1) lies on 0 bisecants : { }
 Off point 237 = P_{247} = (8, 13, 1) lies on 0 bisecants : { }
 Off point 238 = P_{248} = (9, 13, 1) lies on 2 bisecants : { 3, 21 }
 Off point 239 = P_{249} = (10, 13, 1) lies on 2 bisecants : { 14, 22 }
 Off point 240 = P_{250} = (11, 13, 1) lies on 2 bisecants : { 11, 24 }
 Off point 241 = P_{251} = (12, 13, 1) lies on 2 bisecants : { 1, 8 }
 Off point 242 = P_{252} = (13, 13, 1) lies on 3 bisecants : { 0, 9, 23 }
 Off point 243 = P_{254} = (15, 13, 1) lies on 0 bisecants : { }
 Off point 244 = P_{255} = (16, 13, 1) lies on 0 bisecants : { }
 Off point 245 = P_{256} = (0, 14, 1) lies on 4 bisecants : { 2, 3, 23, 25 }
 Off point 246 = P_{257} = (1, 14, 1) lies on 0 bisecants : { }
 Off point 247 = P_{258} = (2, 14, 1) lies on 0 bisecants : { }
 Off point 248 = P_{259} = (3, 14, 1) lies on 2 bisecants : { 16, 19 }
 Off point 249 = P_{260} = (4, 14, 1) lies on 2 bisecants : { 10, 15 }
 Off point 250 = P_{261} = (5, 14, 1) lies on 2 bisecants : { 0, 6 }
 Off point 251 = P_{262} = (6, 14, 1) lies on 0 bisecants : { }
 Off point 252 = P_{263} = (7, 14, 1) lies on 2 bisecants : { 18, 22 }
 Off point 253 = P_{264} = (8, 14, 1) lies on 2 bisecants : { 13, 14 }
 Off point 254 = P_{265} = (9, 14, 1) lies on 2 bisecants : { 21, 27 }
 Off point 255 = P_{266} = (10, 14, 1) lies on 0 bisecants : { }
 Off point 256 = P_{267} = (11, 14, 1) lies on 3 bisecants : { 4, 8, 20 }
 Off point 257 = P_{268} = (12, 14, 1) lies on 0 bisecants : { }
 Off point 258 = P_{269} = (13, 14, 1) lies on 0 bisecants : { }
 Off point 259 = P_{270} = (14, 14, 1) lies on 2 bisecants : { 5, 11 }
 Off point 260 = P_{271} = (15, 14, 1) lies on 2 bisecants : { 17, 24 }
 Off point 261 = P_{272} = (16, 14, 1) lies on 3 bisecants : { 1, 9, 26 }
 Off point 262 = P_{273} = (0, 15, 1) lies on 2 bisecants : { 2, 11 }
 Off point 263 = P_{274} = (1, 15, 1) lies on 2 bisecants : { 16, 27 }
 Off point 264 = P_{275} = (2, 15, 1) lies on 3 bisecants : { 4, 9, 24 }
 Off point 265 = P_{276} = (3, 15, 1) lies on 2 bisecants : { 1, 10 }
 Off point 266 = P_{277} = (4, 15, 1) lies on 4 bisecants : { 19, 20, 22, 23 }
 Off point 267 = P_{278} = (5, 15, 1) lies on 0 bisecants : { }
 Off point 268 = P_{279} = (6, 15, 1) lies on 2 bisecants : { 14, 15 }
 Off point 269 = P_{280} = (7, 15, 1) lies on 0 bisecants : { }
 Off point 270 = P_{281} = (8, 15, 1) lies on 3 bisecants : { 3, 17, 18 }
 Off point 271 = P_{282} = (9, 15, 1) lies on 3 bisecants : { 6, 21, 26 }
 Off point 272 = P_{283} = (10, 15, 1) lies on 2 bisecants : { 8, 13 }
 Off point 273 = P_{284} = (11, 15, 1) lies on 0 bisecants : { }
 Off point 274 = P_{285} = (12, 15, 1) lies on 0 bisecants : { }
 Off point 275 = P_{286} = (13, 15, 1) lies on 0 bisecants : { }
 Off point 276 = P_{287} = (14, 15, 1) lies on 3 bisecants : { 0, 5, 25 }

Off point 277 = $P_{288} = (15, 15, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 278 = $P_{289} = (16, 15, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 279 = $P_{290} = (0, 16, 1)$ lies on 2 bisecants : $\{ 2, 7 \}$
 Off point 280 = $P_{291} = (1, 16, 1)$ lies on 3 bisecants : $\{ 7, 17, 22 \}$
 Off point 281 = $P_{292} = (2, 16, 1)$ lies on 3 bisecants : $\{ 7, 10, 26 \}$
 Off point 282 = $P_{293} = (3, 16, 1)$ lies on 2 bisecants : $\{ 7, 11 \}$
 Off point 283 = $P_{294} = (4, 16, 1)$ lies on 2 bisecants : $\{ 7, 14 \}$
 Off point 284 = $P_{295} = (5, 16, 1)$ lies on 3 bisecants : $\{ 7, 9, 19 \}$
 Off point 285 = $P_{296} = (6, 16, 1)$ lies on 3 bisecants : $\{ 0, 7, 24 \}$
 Off point 286 = $P_{297} = (7, 16, 1)$ lies on 2 bisecants : $\{ 1, 7 \}$
 Off point 287 = $P_{298} = (8, 16, 1)$ lies on 3 bisecants : $\{ 7, 15, 23 \}$
 Off point 288 = $P_{299} = (9, 16, 1)$ lies on 4 bisecants : $\{ 7, 8, 18, 21 \}$
 Off point 289 = $P_{300} = (10, 16, 1)$ lies on 3 bisecants : $\{ 4, 7, 27 \}$
 Off point 290 = $P_{301} = (11, 16, 1)$ lies on 2 bisecants : $\{ 7, 25 \}$
 Off point 291 = $P_{302} = (12, 16, 1)$ lies on 2 bisecants : $\{ 7, 13 \}$
 Off point 292 = $P_{303} = (13, 16, 1)$ lies on 2 bisecants : $\{ 6, 7 \}$
 Off point 293 = $P_{304} = (14, 16, 1)$ lies on 3 bisecants : $\{ 5, 7, 20 \}$
 Off point 294 = $P_{306} = (16, 16, 1)$ lies on 3 bisecants : $\{ 3, 7, 16 \}$

The gradient

The gradient of the quartic curve is :

$$13X_0^3 + 11X_1^3 + 11X_2^3 + 6X_0^2X_1 + 16X_0^2X_2 + 10X_0X_1^2 + 2X_1^2X_2 + 4X_0X_2^2 + 11X_0X_1X_2$$

$$(13, 11, 11, 6, 16, 10, 2, 4, 0, 11)$$

$$2X_0^3 + X_1^3 + 9X_2^3 + 10X_0^2X_1 + 14X_0^2X_2 + 16X_0X_1^2 + 9X_1^2X_2 + 4X_1X_2^2 + 4X_0X_1X_2$$

$$(2, 1, 9, 10, 14, 16, 9, 0, 4, 4)$$

$$11X_0^3 + 3X_1^3 + 9X_2^3 + 14X_0^2X_1 + 4X_0^2X_2 + 2X_0X_1^2 + 4X_1^2X_2 + 16X_0X_2^2 + 10X_1X_2^2$$

$$(11, 3, 9, 14, 4, 2, 4, 16, 10, 0)$$